

17-2b Oligopolies as a Prisoners' Dilemma

What does the prisoners' dilemma have to do with markets and imperfect competition? It turns out that the game oligopolists play in trying to reach the monopoly outcome is similar to the game that the two prisoners play in the prisoners' dilemma.

Consider again the choices facing Jack and Jill. After prolonged negotiation, the two suppliers of water agree to keep production at **30** gallons so that the price will be high and together they will earn the maximum profit. After they agree on production levels, however, each of them must decide whether to cooperate and honor this agreement or to ignore it and produce at a higher level. [Figure 2](#) shows how the profits of the two producers depend on the strategies they choose.

Figure 2

Jack and Jill's Oligopoly Game

In this game between Jack and Jill, the profit that each earns from selling water depends on both the quantity he or she chooses to sell and the quantity the other chooses to sell.

Suppose you are Jack. You might reason as follows: "I could keep production at **30** gallons as we agreed, or I could raise my production and sell **40** gallons. If Jill lives up to the agreement and keeps her production at **30** gallons, then I earn a profit of **\$2,000** by selling **40** gallons and **\$1,800** by selling **30** gallons. In this case, I am better off with the higher-level production. If Jill fails to live up to the agreement and produces **40** gallons, then I earn **1,600** by selling **40** gallons and **\$1,500** by selling **30** gallons. Once again, I am better off with higher production. So, regardless of what Jill chooses to do, I am better off reneging on our agreement and producing at the higher level."

Producing 40 gallons is a dominant strategy for Jack. Of course, Jill reasons in exactly the same way, and so both produce at the higher level of 40 gallons. The result is the inferior outcome (from Jack and Jill's standpoint) with low profits for each of the two producers.

This example illustrates why oligopolies have trouble maintaining monopoly profits. The monopoly outcome is jointly rational, but each oligopolist has an incentive to cheat. Just as self-interest drives the prisoners in the prisoners' dilemma to confess, self-interest makes it hard for the oligopolists to maintain the cooperative outcome with low production, high prices, and monopoly profits.

Case Study: OPEC and the World Oil Market

Our story about the town's market for water is fictional, but if we change water to crude oil, and Jack and Jill to Iran and Iraq, the story is close to reality. Much of the world's oil is produced by a few countries, mostly in the Middle East. These countries together make up an oligopoly. Their decisions about how much oil to pump are much the same as Jack and Jill's decisions about how much water to pump.

The countries that produce much of the world's oil have formed a cartel, called the Organization of Petroleum Exporting Countries (OPEC). Originally formed in 1960, OPEC now includes Saudi Arabia, Iraq, Iran, United Arab Emirates, Kuwait, Venezuela, and several other nations. Together, OPEC countries control about 80 percent of the world's oil reserves. Like any cartel, OPEC tries to raise the price of its product through a coordinated reduction in quantity produced. OPEC tries to set production levels for each of the member countries.

The problem that OPEC faces is much the same as the problem that Jack and Jill face in our story. The OPEC countries would like to maintain a high price for oil. But each member of the cartel is tempted to increase its production to get a larger share of the total profit. OPEC members frequently agree to reduce production but then cheat on their agreements.

OPEC was most successful at maintaining cooperation and high prices in the period from 1973 to 1985. The price of crude oil rose from \$3 a barrel in 1972 to \$11 in 1974 and then to \$35 in 1981. But in the mid-1980s, member countries began arguing about production levels, and OPEC became ineffective at maintaining cooperation. By 1986 the price of crude oil had fallen back to \$13 a barrel.

In recent years, the members of OPEC have continued to meet regularly, but they have been less successful at reaching and enforcing agreements. Changes in technology, such as the development of fracking, have expanded oil supply around the world and reduced OPEC's market power. As a result, fluctuations in oil prices have been driven more by the natural forces of supply and demand than by the cartel's artificial restrictions on production.

